

flyPad descent calculator

The A380 OIS IN-FLT PERF DES module on the flyPad of both aircraft, computed with the FMS performance model

Branch	pr/24-efb-descent-calculator
Aircraft	A380X and A32NX (flyPad, A380 FMC, A320 MCDU)
Commits	469b9a0c8 feat(efb): descent calculator in place of the top of descent calculator; 2026-09-29: d39193281 feat(efb): T/D distance slider, buttons at the bottom
Base	pr/23-efb-landing-calculator (FMS data answers, performance store and texts)
Status	CI checks pass; merged into develop 99f00b531. Checked in the flyPad harness; in-sim test to do.

Purpose

The flyPad Top of Descent tab only applied a descent angle to a ground speed. The real A380 computes descents in the OIS IN-FLT PERF DES module (A380 FCOM PER-IFT-DES), with the aircraft performance, speeds and conditions.

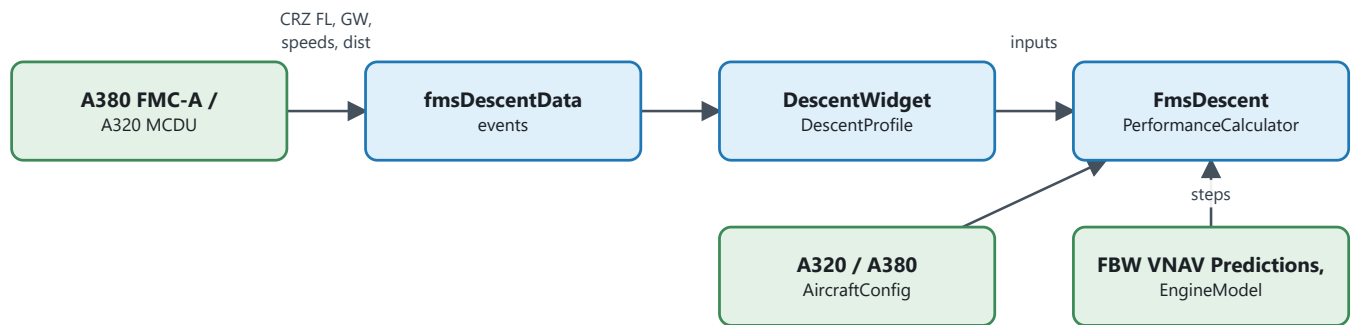
Specification (A380 FCOM PER-IFT-DES, PER-IFT-DES-STD; A320 FCOM PER-DES-STD)

Part	Content
Descent type	ECON (FMS managed Mach / CAS), STANDARD (A380 M.85 / 300 / 250 kt below FL100, A320 M.78 / 300 / 250), GIVEN V/S, EMERGENCY (MMO / VMO with the speed brakes)
Descent	initial altitude (A/C), target altitude, speed brakes
Speed schedule	MACH, SPD, SPD LIM, LIM ALT; empty = the standard schedule
Conditions	gross weight, ISA deviation (A/C from the SAT), wind component, anti-ice (off / engine / total), fuel factor
FMS import	cruise level, destination elevation + 1500 ft, gross weight, speed limit, ECON speeds, distance to the destination
RESULTS	time, distance, fuel, mean V/S and FPA, 3 x altitude rule; crossover and deceleration
Profile	chart (T/D, CROSSOVER, DECEL, SPD LIM, the aircraft) or table every 5000 ft (time, distance, fuel, CAS, Mach, TAS, V/S, FPA)
DESCENT CHECK	distance to the target vs the profile: start in X NM (Y min), on profile, or late with the V/S and FPA needed

Model

- The FBW VNAV prediction steps (Predictions.altitudeStep, verticalSpeedStep, speedChangeStep, idle N1 of the engine model) with the A320 / A380 aircraft config, integrated in 1000 ft steps, so the flyPad agrees with the FMS descent predictions.
- Deceleration to the speed limit on the -1 degree idle path of the FMS descent strategy, ending at the limit altitude.
- GIVEN V/S steeper than the idle descent: idle there, with a warning (the pressure altitude below which it applies).
- Anti-ice: the corrections of the A320 FCOM descent table, marked as estimates (amber).
- Check (A320, M.78 / 300 / 250, 65 t, FL390 to 1500 ft): FCOM 106 NM / 17.4 min / 165 kg, calculator 114 NM / 18.8 min / 146 kg (the FCOM table is for the CFM56 A320, the model is the A320neo).

 Component added or changed by the feature  MSFS / simulator  Aircraft system (systems host, FMS, FWS)  Display / page
 Data, LVar, storage  Crew, flypad, external program



Architecture

Interfaces

Name	Type	Meaning
fms_descent_data_request	event bus	the flyPad asks the FMS for its descent data (request id)
fms_descent_data	event bus	cruise level, destination and elevation, distance, gross weight, cost index, ECON speeds, speed limit

Tests

3 unit tests in `FmsDescentPerformanceCalculator.spec.ts`: close to the A320 FCOM table; wind, temperature and anti-ice; EMERGENCY and the given V/S check.

Limitations

- The deceleration to 250 kt takes about 5000 ft, as in the FBW FMS model (a -1 degree idle path).
- No bleed in the engine model: the anti-ice effect comes from the A320 FCOM table on both aircraft.
- The old TODCalculator components are removed; its store slice is left in place, unused.

T/D distance slider (2026-09-29)

- Moves the start of the descent from 60 NM later (half the distance at most) to 60 NM earlier than the calculated T/D; reversed like the chart (farther on the left).
- At the mean ground speed of the calculated descent: $V/S = \text{mean rate} \times D / D'$, $\tan \text{FPA} = \tan \text{mean FPA} \times D / D'$.
- Later: steeper than the idle descent (amber: speed brakes or a higher speed; red when the calculation already has the speed brakes). Earlier: shallower (thrust, more fuel). Given V/S descents: the figures only.
- The moved T/D is drawn dashed, straight to the target. Calculate and Clear sit at the bottom of the column, as on the takeoff and landing pages.

PR description (from the PR draft)

From `pr-drafts/24-efb-descent-calculator.md`, section "Summary of Changes" onwards.

Summary of Changes

The Top of Descent tab of the flyPad performance page becomes a descent calculator, after the A380 OIS IN-FLT PERF DES module (A380 FCOM PER-IFT-DES), on both aircraft.

- **Descent types:** ECON (the managed descent Mach and CAS of the FMS), STANDARD (the FCOM speed schedule: A380 M.85 / 300 kt / 250 kt below FL100, PER-IFT-DES-STD; A320 M.78 / 300 kt / 250 kt, PER-DES-STD), GIVEN V/S, EMERGENCY (MMO / VMO with the speed brakes).
- **Inputs:** initial altitude (A/C: the current one) and target altitude; the speed schedule (MACH, SPD, SPD LIM, LIM ALT; empty = the standard one); gross weight, ISA deviation (A/C: from the current SAT), wind component, anti-ice (off / engine / total), fuel factor, speed brakes. **Fill data from FMS:** cruise level (or the current altitude when below), destination elevation + 1500 ft, gross weight, the FMS speed limit, the ECON speeds and the distance to the destination.
- **RESULTS:** time, distance, fuel, mean V/S and flight path angle, and the 3 x altitude rule for comparison; the crossover altitude and the deceleration to the speed limit.
- **Profile:** a chart of altitude against the distance to the target (T/D, crossover, DECEL, SPD LIM) or a table every 5000 ft (time, distance, fuel, CAS, Mach, TAS, V/S, FPA), like the FCOM results table.
- **DESCENT CHECK:** with the distance to the target, the aircraft altitude and ground speed: start the descent in X NM (Y min), on profile, or N NM late with the V/S and FPA needed; the aircraft is drawn on the profile.
- **Model:** the FBW VNAV prediction steps (`Predictions.altitudeStep`, `verticalSpeedStep`, `speedChangeStep`, with the idle N1 of the engine model) and the aircraft config of each aircraft, integrated in 1000 ft steps; the deceleration to the speed limit on the -1 degree idle path of the FMS descent strategy, so the flyPad agrees with the FMS predictions. GIVEN V/S steeper than idle: idle there, with a warning. Anti-ice: the corrections of the A320 FCOM descent table, shown as estimates (amber).
- **FMS import:** the A380 FMC-A and the A320 MCDU answer the flyPad (shared `fmsDescentData` events).
- The old `TODCalculator` components are removed (its `todCalculator` store slice is left in place; nothing reads it any more).
- 3 unit tests (close to the A320 FCOM descent table; wind, temperature and anti-ice; EMERGENCY and the given V/S check).

Check against the A320 FCOM descent table (M.78 / 300 / 250, 65 t, FL390 to 1500 ft): FCOM 106 NM / 17.4 min / 165 kg, calculator 114 NM / 18.8 min / 146 kg. The FCOM table is for the CFM56 A320; the FBW A320neo model descends about 8 % longer.

- **T/D distance slider** under the profile: moves the start of the descent up to 60 NM earlier or later than the calculated T/D (reversed like the chart: farther on the left), with the V/S and FPA it needs at the mean ground speed of the calculated descent, against the calculated ones; later = steeper than the idle descent (speed brakes or a higher speed, red when the calculation already has the speed brakes), earlier = shallower (thrust, more fuel); the moved T/D is drawn dashed to the target. The Calculate and Clear buttons are at the bottom of the column, as on the takeoff and landing pages.

Cockpit API Changes

None (two new event bus topics, `fms_descent_data_request` and `fms_descent_data`).

Screenshots (if necessary)

TO ADD: the A380 screen with the profile chart and the table, the A320 screen, the DESCENT CHECK late and early.

References

- A380 FCOM PER-IFT-DES (IN-FLT PERF DES module, descent types, speed profile, conditions), PER-IFT-DES-STD.

- A320 FCOM PER-DES-STD (standard descent table and its anti-ice corrections).

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. flyPad > Performance > Top of Descent, in cruise with a flight plan: Fill data from FMS: the cruise level, target, weight, speed limit and distance are filled. CALCULATE: RESULTS and the profile.
2. ECON: the speeds of the FMS; compare the T/D with the one of the ND.
3. GIVEN V/S 3000 ft/min, EMERGENCY: shorter descents; a V/S steeper than idle gives the warning.
4. DESCENT CHECK: start in X NM before the T/D; late after it, with the V/S and FPA needed.
5. `npx vitest run fbw-a32nx/src/systems/fmgc/src/performance/FmsDescentPerformanceCalculator.spec.ts` (3 tests).